

image of at most  $k!$  injective homomorphisms  $H \rightarrow G$ , so

$$h(G) \leq k! \cdot d \|G\| \leq k! \cdot d \cdot n^2 \leq \eta n_0 \cdot n^2 \leq \eta n^3.$$

The removal lemma therefore yields a set  $E$  of at most  $\eta n^2$  edges in  $G$  such that  $H \not\subseteq G - E$ . Like every edge of  $G$ , each of the edges in  $E$  lies in at most  $d$  copies of  $H$  in  $G$ . So we also have

$$h(G) \leq k! \cdot d |E|.$$

As the copies of  $H$  are thinly spread over  $G$ , every edge of  $G$  lies in the image of some injective homomorphism  $H \rightarrow G$ . Hence  $h(G) \geq \|G\|/m$ , so

$$\|G\| \leq m h(G) \leq m k! d |E| \leq m k! d \eta n^2 = \delta n^2.$$

As the number  $h$  of copies of  $H$  in  $G$  is no bigger than  $h(G)$ , the above inequality also shows  $h \leq h(G) \leq \delta n^2$ , as claimed.  $\square$

[7.7.3] **Corollary 7.6.5.** *For every  $\delta > 0$  there exists an integer  $n_0$  such that the following holds for every graph  $G$  of order  $n \geq n_0$ : if every edge of  $G$  lies in exactly one triangle, then  $G$  has at most  $\delta n^2$  edges.*  $\square$

## 7.7 Szemerédi's theorem

When Szemerédi proved his first version of the regularity lemma, he did not think of it as a precursor to the fundamental structure theorem for all of extremal graph theory that it has since become. He thought of it, well, as a lemma: a lemma he needed in a proof of a conjecture of Erdős and Turán about ‘arithmetic progressions’ of integers. We shall describe this theorem in this section, and then use the tools from Section 7.6 to derive its simplest case in just over a page.

arithmetic  
progression

length An *arithmetic progression* is an increasing sequence of integers of the form  $a, a + d, a + 2d, \dots$ , one in which adjacent terms always differ by the same amount. The number of terms in the sequence is its *length*. One of the earliest results in combinatorics is *van der Waerden's theorem* that, no matter how we partition the natural numbers into finitely many classes, one of these will contain arbitrarily long arithmetic progressions.<sup>18</sup> Szemerédi's theorem explores the deeper reasons why this is so.

upper  
density

In any partition  $\{A_1, \dots, A_p\}$  of  $\mathbb{N}$  into finitely many classes, one of these  $A_i$  will have positive *upper density*: for some  $\delta > 0$ , in our case certainly for  $\delta = 1/p$ , arbitrarily large  $n \in \mathbb{N}$  will satisfy  $|A_i \cap n| \geq \delta n$

<sup>18</sup> Compare Exercise 5 in Chapter 9.